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RESEARCH ARTICLE

Burn Severity Classification Using Convolutional Neural Networks With Thermal Imaging and Diffuse Reflectance Spectroscopy

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ABSTRACT Accurate assessment of burn severity is essential for clinical decision-making and forensic investigations. However, conventional visual examination is subjective and often cannot reliably distinguish between different burn depths. Although machine learning (ML) and deep learning (DL) techniques have improved automated burn classification, most existing studies rely on a single sensing modality and lack synchronized thermal and optical measurements acquired under identical experimental conditions. This study presents a controlled experimental framework for burn severity classification using thermal imaging and Diffuse Reflectance Spectroscopy (DRS). A controlled experimental framework was established using ex vivo porcine skin, where burn severity was systematically induced by varying the exposure time while maintaining a constant temperature. A synchronized dual-modality dataset comprising approximately 4,060 thermal images and 1,368 diffuse reflectance spectra was acquired under identical burn conditions, representing five burn conditions, including unburned tissue and four burn severity levels. Two independent deep learning models were developed and evaluated: a convolutional neural network (CNN) for thermal imaging and a one-dimensional convolutional neural network (1D-CNN) for Diffuse Reflectance Spectroscopy. The thermal imaging model achieved an overall classification accuracy of 97.17%, with precision, recall, and F1-score exceeding 97%, whereas the Diffuse Reflectance Spectroscopy model achieved an average classification accuracy of 92.91% $\pm$ 2.93% using five-fold stratified cross-validation. The results demonstrate that both sensing modalities provide reliable burn severity classification, while thermal imaging achieved higher standalone classification performance. The proposed synchronized dataset and controlled experimental framework provide an objective benchmark for future multimodal fusion methods for burn severity assessment.

INDEX TERMS Burn severity classification, thermal imaging, diffuse reflectance spectroscopy, convolutional neural network, image processing, spectral analysis.

I. INTRODUCTION

Accurate burn severity classification is essential for clinical decision-making, forensic investigations, and treatment planning because it directly influences wound management and patient outcomes [1], [2]. Burn injuries remain a major global health problem, causing approximately 180,000 deaths each year, most of which occur in low- and middle-income countries [3]. Early and accurate assessment of burn depth is therefore critical for selecting appropriate

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treatment, planning surgical intervention, and improving recovery [4], [5].

Burn severity is traditionally assessed through visual examination and manual palpation. However, these methods are subjective and often fail to distinguish between intermediate burn depths, particularly during the early stages of injury [6], [7]. Even experienced burn specialists report diagnostic accuracies of only 60–80% when relying solely on clinical examination, with the greatest uncertainty occurring in partial-thickness burns [8]. These limitations have motivated the development of objective, quantitative, and non-invasive techniques for burn assessment [9], [10].

Infrared thermal imaging (IRT) has become an important non-contact technique for burn assessment because it measures surface temperature distributions associated with tissue perfusion, inflammation, and thermal damage [11], [12]. Superficial burns generally appear warmer because of increased blood flow, whereas deeper burns appear cooler due to vascular destruction and reduced perfusion [13], [14], [15], [16], [17]. Recent advances in machine learning (ML) and deep learning (DL), particularly convolutional neural networks (CNNs), have further improved thermal image analysis by enabling automatic feature extraction and accurate burn classification [18].

Several studies have applied CNN-based methods to classify burn severity using visible-light or thermal images. Although these approaches have shown promising performance, most rely on a single sensing modality and are often trained using heterogeneous clinical datasets acquired under different imaging conditions [19], [20], [21]. Consequently, variations in illumination, skin pigmentation, imaging devices, and annotation quality can reduce model robustness and limit generalization [1], [22], [23].

Diffuse reflectance spectroscopy (DRS) provides complementary information by measuring wavelength-dependent tissue absorption and scattering, which reflect biochemical composition and structural changes caused by thermal injury [24], [25], [26], [27], [28], [29], [30], [31]. Previous studies have shown that DRS can differentiate healthy and burned tissue by detecting changes in chromophores, water content, and collagen structure. However, most studies evaluate thermal imaging and spectroscopy independently, making direct comparison between the two modalities difficult [24], [25].

Controlled experimental burn models provide a reliable framework for generating reproducible burn injuries. Burn severity is commonly controlled by varying the exposure time while maintaining a constant temperature, following the well-established time–temperature relationship of thermal damage [3], [33], [34]. This approach produces consistent burn grades with reliable ground-truth labels, making it well suited for developing and evaluating machine learning models.

To address these challenges, this study establishes a controlled experimental framework using ex vivo porcine skin to collect synchronized thermal images and Diffuse Reflectance Spectroscopy measurements from the same tissue samples under identical burn conditions. Burn severity was systematically varied by changing the exposure time while maintaining a constant thermal load. Approximately 4,060 thermal images and 1,368 diffuse reflectance spectra were collected across five burn conditions. A CNN was developed to classify burn severity from thermal images, while a one-dimensional CNN (1D-CNN) was trained using the spectroscopic data. The two models were evaluated independently to enable an objective comparison of the two sensing modalities. Although multimodal fusion is beyond the scope of this work, the synchronized dataset provides a foundation for future fusion-based approaches.

The main contributions of this work are summarized as follows.

1. A controlled experimental framework was established to acquire synchronized thermal images and diffuse reflectance spectra from the same porcine skin specimens under identical burn conditions.
2. A well-balanced dual-modality dataset comprising approximately 4,060 thermal images and 1,368 diffuse reflectance spectra was created for five burn severity levels.
3. A CNN and a 1D-CNN were developed and evaluated independently for burn severity classification using thermal images and spectroscopic data, respectively.
4. A comparative analysis was performed to evaluate the strengths and limitations of thermal imaging and DRS for burn severity classification.
5. The proposed controlled experimental framework and synchronized dual-modality dataset provide a benchmark for future multimodal burn assessment methods.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the experimental methods and machine learning framework. Section IV presents the experimental results and discussion. Finally, Section V concludes the paper and outlines future research directions.

II. RELATED WORK

Accurate burn severity classification has been extensively studied because it is essential for diagnosis, treatment planning, and patient recovery. To improve the accuracy and objectivity of burn assessment, researchers have investigated different sensing techniques and artificial intelligence methods. This section reviews previous studies on thermal imaging, DRS, artificial intelligence methods, and multimodal approaches.

A. BURN SEVERITY ASSESSMENT USING THERMAL IMAGING

IRT has become an effective non-invasive technique for burn assessment because it measures temperature variations associated with tissue perfusion and viability [11], [12]. Superficial burns usually show higher temperatures due to increased blood flow, whereas deeper burns appear cooler because thermal injury damages the microvasculature and reduces tissue perfusion [39], [40].

Several studies have investigated the use of thermal imaging for burn assessment. Hardwicke et al. and Renkielska et al. showed that both static and active thermography can distinguish superficial and deep burns by analyzing temperature distribution and thermal recovery patterns [41], [42]. Singer et al. further reported that thermal imaging improves burn assessment compared with visual examination and supports more informed clinical decisions [43].

Despite these advantages, thermal imaging has several limitations. Image quality is sensitive to environmental conditions such as room temperature, humidity, and airflow, making standardized acquisition procedures essential [44]. In addition, temperature measurements mainly describe surface conditions and do not directly reflect the biochemical and structural changes caused by thermal injury [24]. In clinical practice, thermal images are often interpreted visually, which may introduce observer variability and reduce the reproducibility of burn assessment. To overcome these limitations, recent studies have integrated thermal imaging with artificial intelligence methods to improve classification accuracy and reduce subjective interpretation [45]. Comprehensive reviews have also emphasized the need for standardized imaging protocols and automated analysis for reliable clinical implementation [11], [27].

B. BURN SEVERITY ASSESSMENT USING OPTICAL SPECTROSCOPY

DRS provides complementary information by measuring tissue absorption and scattering, which change as a result of thermal damage [26], [30]. The measured spectra reflect variations in tissue composition, including oxyhemoglobin, deoxyhemoglobin, melanin, collagen, and water content, providing physiological information that cannot be obtained from temperature measurements alone [31], [39].

Previous studies have demonstrated the potential of DRS for burn assessment. Sowa et al. used near-infrared spectroscopy to classify burn severity in porcine skin with high accuracy [39]. Cross et al. reported good agreement between spectroscopic measurements and histopathological findings while highlighting the influence of tissue variability on classification performance [16]. Other researchers improved tissue characterization by estimating optical properties through diffusion-based models and extended spectral analysis [26], [30], [31].

Although DRS provides detailed physiological information, it also has practical limitations. Most systems acquire measurements from a relatively small tissue area, resulting in limited spatial coverage [46]. In addition, tissue optical properties vary with anatomical location, hydration, age, and skin pigmentation, making robust model development more challenging [47].

TABLE 1. Comparison of representative studies in burn severity classification.

Study	Modality	Burn Model	Classification	Key limitation
Suha & Sanam [19].	Visible-light images	Clinical (uncontrolled)	Deep CNN	Single modality; uncontrolled conditions
Pabitha & Vanathi [20].	Visible-light images	Clinical (small dataset)	DenseMask RCNN	Small dataset; no standardized protocol
Rahman et al. [21].	Visible-light images	Clinical	Multiple DL architectures	Single modality; no physiological data
Sowa et al. [39].	NIR spectroscopy	Porcine (controlled)	Statistical classification	Spectroscopy only; no imaging
Cross et al. [16].	NIR spectroscopy	Clinical	Spectral analysis	Point measurement; no thermal data
Renkielska et al. [42].	Active dynamic thermography	Porcine (controlled)	Thermal recovery analysis	Thermal only; no spectroscopy
Lertsakdadet et al. [24].	SFDI + LSI (optical)	Clinical	Perfusion mapping	No thermal imaging or DRS
Burmeister et al. [23].	SFDI + LSI	Porcine (controlled)	Optical parameter analysis	No thermal imaging; limited ML
This work	**Thermal imaging + DRS**	**Porcine (controlled, graded)**	**CNN + ML**	**Ex vivo model; future fusion needed**

C. ARTIFICIAL INTELLIGENCE FOR BURN CLASSIFICATION

Artificial intelligence has significantly improved automated burn severity classification. Early studies relied on handcrafted image features, such as color, texture, and shape, combined with conventional classifiers [22]. While these methods showed promising results, their performance depended heavily on feature design and image quality.

Recent studies have increasingly adopted CNNs because they automatically learn discriminative features directly from raw data [36], [48], [49]. Suha and Sanam developed a CNN model for burn classification using skin images [19], while Pabitha and Vanathi proposed DenseMask RCNN for simultaneous burn segmentation and severity grading [20]. Rahman et al. compared several deep learning architectures and demonstrated the effectiveness of CNN-based models for burn classification [21].

Transfer learning has also been widely applied to compensate for the limited size of medical datasets by adapting models pretrained on large image collections [50], [51]. However, the characteristics of natural images differ considerably from thermal and biomedical data, which may limit the transferability of learned features [52].

Deep learning has also been successfully applied to spectral analysis. CNNs have also been applied successfully to one-dimensional optical signals, demonstrating that convolutional feature extraction is effective for both image and spectral data [53], [54]. These findings support the use of a 1D-CNN for burn classification using spectral measurements.

Although these studies have achieved encouraging results, several challenges remain. Most existing approaches rely on a single sensing modality, use heterogeneous datasets collected under different experimental conditions, or lack standardized burn induction protocols. These limitations make objective comparison between sensing techniques difficult and reduce the robustness and reproducibility of developed models [1], [7], [9]. Furthermore, the limited availability of synchronized multimodal datasets has constrained the development and evaluation of multimodal deep learning methods for burn severity classification [23], [24], [25].

D. MULTIMODAL APPROACHES AND RESEARCH GAP

Multimodal sensing has gained increasing attention because different sensing techniques provide complementary information about tissue condition [55], [56]. Thermal imaging provides spatial temperature information, whereas Diffuse Reflectance Spectroscopy captures physiological and structural changes within the tissue [16], [24]. Combining these complementary measurements has the potential to improve burn severity classification.

Recent studies have also demonstrated that convolutional neural networks can effectively learn features from both images and one-dimensional signals in applications beyond medical imaging, including optical fiber sensing [53], [54]. These studies highlight the flexibility of deep learning architectures for analyzing different types of sensing data and support their application to burn severity classification.

Despite the potential of multimodal sensing, synchronized datasets remain limited. Most existing studies acquired thermal and spectroscopic measurements separately, often using different tissue samples, acquisition times, or experimental protocols. Consequently, direct comparison between sensing modalities is difficult, and the development of multimodal learning methods remains limited [25]. Existing multimodal studies have primarily combined optical imaging techniques, such as SFDI and LSI, without integrating thermal imaging and Diffuse Reflectance Spectroscopy within the same controlled experimental framework [23], [24], [25].

As summarized in Table 1, existing studies have demonstrated promising results for burn severity classification using thermal imaging, Diffuse Reflectance Spectroscopy, or artificial intelligence methods. However, most previous studies relied on a single sensing modality, heterogeneous datasets, or different experimental protocols, while synchronized multimodal datasets acquired under identical burn conditions remain scarce. Consequently, objective comparison between sensing techniques and the development of multimodal learning methods remain challenging. Without synchronized acquisition from the same tissue samples under identical burn conditions, direct cross-modality comparison and future multimodal fusion cannot be performed reliably. To the best of our knowledge, no previous study has reported such a synchronized dataset. To address this limitation, the present study establishes a controlled experimental framework to acquire synchronized thermal images and Diffuse Reflectance Spectroscopy measurements and develops separate deep learning models for each sensing modality. This approach enables an objective comparison of their classification performance and provides a benchmark for future multimodal fusion research.

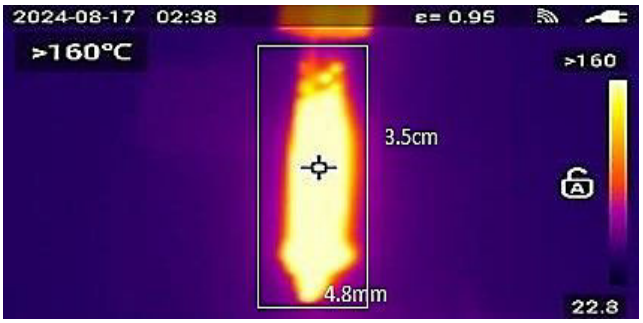


FIGURE 1. Experimental burn induction apparatus (incinerator) used in the study, including the cylindrical brass heating probe (3.5 cm length and 4.8 mm diameter) employed for controlled burn induction.

III. MATERIALS AND METHODS

This section describes the controlled experimental framework developed for automated burn severity classification. It presents the preparation of the ex vivo porcine skin samples, the burn induction protocol, the experimental setup, synchronized acquisition of thermal images and Diffuse Reflectance Spectroscopy measurements, dataset preparation, and the deep learning models developed for each sensing modality. The proposed framework was designed to provide standardized burn induction, synchronized data acquisition, and reproducible experimental conditions, enabling an objective comparison between thermal imaging and Diffuse Reflectance Spectroscopy.

A. PORCINE SKIN SAMPLE

Fresh ex vivo porcine skin samples were harvested from the paravertebral region and cut into pieces measuring 30 × 20 mm with a thickness of 2.0 ± 0.2 mm. Before the experiments, the samples were equilibrated at 22 ± 1 °C and hydrated in 0.9% NaCl saline to maintain approximately 70% physiological water content, thereby minimizing spectral drift and changes in surface emissivity during data acquisition.

Porcine skin is widely recognized as an appropriate model for human skin because of its similar epidermal thickness, dermal structure, collagen composition, barrier function, and immunohistochemical characteristics [38], [57]. It also exhibits comparable optical absorption and scattering

properties in the visible and near-infrared wavelength ranges, supporting its use in both thermal imaging and DRS studies [58]. These similarities make porcine skin a reliable model for investigating burn-induced thermal and optical changes under controlled experimental conditions.

Despite these advantages, ex vivo porcine skin has inherent limitations. The absence of blood perfusion, inflammatory response, and tissue healing prevents complete replication of physiological conditions observed in living tissue. Consequently, caution should be exercised when translating the findings directly to clinical applications [59]. Nevertheless, the burn severity patterns and optical responses observed in porcine skin are consistent with those reported in previous studies, supporting its suitability for burn research as well as future clinical and forensic investigations [60], [61].

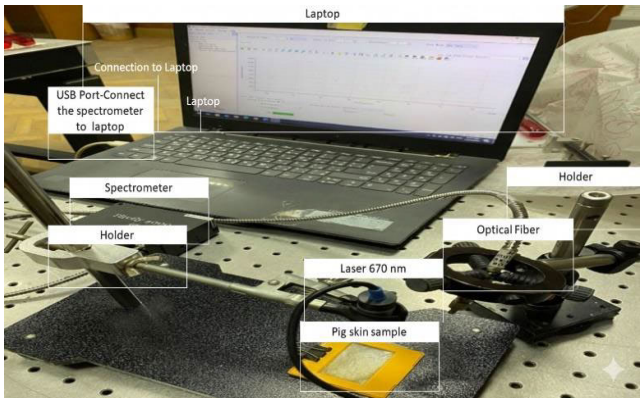


FIGURE 2. The experimental setup for laser irradiation and capturing the thermal image and diffuse reflectance.

B. BURN INDUCTION PROTOCOL

Burn severity was controlled by varying the contact duration while maintaining a constant thermal load, following the time–temperature relationship of thermal tissue injury. Four burn severity levels were defined based on the exposure time:

- Level 1 (4 s): Superficial epidermal damage.
- Level 2 (6 s): Partial-thickness injury.
- Level 3 (10 s): Deep partial-thickness injury.
- Level 4 (15 s): Full-thickness tissue necrosis with structural denaturation.

Burn injuries were induced using the custom-designed heating element shown in Figure 1. The heating element was maintained at a stabilized temperature of 160 °C and had a length of 3.5 cm and a diameter of 4.8 mm. The exposure time was varied while all other experimental conditions remained constant, enabling reproducible generation of the four burn severity levels across all specimens.

To ensure consistent thermal contact, the cylindrical brass heating probe (3.5 cm length and 4.8 mm diameter) was mounted on a mechanical stand that applied a constant force of 200 g. The probe was positioned perpendicular to the skin surface to maintain uniform contact during burn induction. Before each experiment, the skin surface was cleaned and dried, and the probe temperature was verified at 160 °C using an integrated thermocouple. The contact duration was controlled using a digital timer to ensure accurate and repeatable burn induction. Minor variations in contact pressure caused by tissue heterogeneity and surface curvature were considered unavoidable experimental variability and contributed to the diversity of the acquired dataset.

The burn severity levels were defined according to established protocols based on controlled exposure time at 160 °C. However, histopathological analysis was not performed to verify the corresponding burn depths, which represents a limitation of the present study. Nevertheless, the adopted protocol is consistent with previously reported porcine burn models [62], in which porcine skin serves as a reliable surrogate for human skin because of its similar anatomical and thermal characteristics. Future work will incorporate histological analysis to establish a direct correlation between the applied exposure parameters and the resulting burn depth.

C. EXPERIMENTAL SETUP

The experimental setup, shown in Figure 2, was developed to induce controlled thermal injury and perform synchronized thermal imaging and DRS measurements under standardized experimental conditions. The system was designed to provide reproducible burn induction, high measurement accuracy, and synchronized data acquisition for subsequent burn severity classification.

A continuous-wave laser operating at a wavelength of 670 nm was used as the excitation source. The porcine skin samples were placed in a precision-machined specimen holder to maintain a constant focal distance and ensure consistent positioning throughout the experiments. This configuration minimized measurement variability and improved the repeatability of both thermal and optical data acquisition.

The measurements were performed to monitor time-dependent changes in tissue diffuse reflectance at a fixed wavelength during laser irradiation. Therefore, the DRS system was intentionally designed to operate at 670 nm rather than across multiple wavelengths. The objective was to monitor changes in diffuse reflectance resulting from laser-induced alterations in tissue absorption and scattering, including variations in tissue hydration, refractive-index matching, and structural integrity. Consequently, this study evaluates the dynamic optical response of thermally damaged tissue at 670 nm instead of its wavelength-dependent spectral characteristics.

To obtain complementary information from the same tissue sample, the experimental setup integrates two synchronized sensing modalities: thermal monitoring and optical characterization. Thermal monitoring provides real-time surface temperature distribution during and after burn induction, whereas optical characterization measures changes in tissue absorption and scattering associated with thermal damage. The instrumentation and data acquisition procedures for each sensing modality are described in the following subsections.

D. THERMAL MONITORING

Thermal monitoring was performed using a high-sensitivity infrared thermal camera (UTi120TN, UNI-T, China) to capture the surface temperature distribution of the porcine skin samples during and after burn induction. The non-contact imaging approach enables continuous monitoring of tissue temperature while avoiding disturbance to the sample throughout the measurement process.

The camera employs an uncooled focal-plane array (UFPA) detector with a native infrared resolution of 120 × 90 pixels, a pixel pitch of 17 μm, and an operating spectral range of 7.5–14 μm. The optical system provides a 50° × 38° field of view (FOV), an instantaneous field of view (IFOV) of 7.6 mrad, a focal length of 2.28 mm, and an aperture of F1.13 with a fixed-focus configuration. Thermal images were acquired at a frame rate of 25 Hz, enabling continuous monitoring of the temperature evolution during burn induction.

The camera supports temperature measurement ranges of −20 to 150 °C and 100 to 400 °C, with automatic range switching and an adjustable emissivity setting from 0.01 to 1.00 to accommodate different materials. According to the manufacturer's specifications, the measurement accuracy is ±2 °C or ±2% of the reading (whichever is greater) for temperatures of 0 °C or higher, and ±5 °C or ±5% for temperatures below 0 °C, under ambient conditions of 5–30 °C and a target distance of 0.5–3 m. The thermal sensitivity (NETD) is specified as ≤50 mK (0.05 °C).

To ensure consistent measurements, the camera was positioned at a fixed distance from the sample throughout all experiments. This configuration maintained a constant spatial resolution and minimized measurement variability between successive acquisitions. The recorded thermal images were subsequently used to develop and evaluate the thermal image classification model.

E. OPTICAL CHARACTERIZATION

Optical characterization was performed using DRS to evaluate the optical response of thermally damaged tissue. Unlike thermal imaging, which measures surface temperature, DRS characterizes changes in tissue absorption and scattering caused by thermal injury. These optical variations provide complementary information about tissue composition and structural changes, supporting more comprehensive burn severity assessment.

The DRS measurements were acquired using a fiber-optic spectrometer (Aurora 4000, Changchun New Industries Optoelectronics Tech. Co., Ltd., China). A fiber-optic probe was positioned perpendicular to the tissue surface to collect diffusely reflected light from the irradiated region. The measurement geometry was maintained throughout all experiments to ensure consistent optical coupling and improve the repeatability of the acquired signals.

The DRS system operated at a fixed wavelength of 670 nm during laser irradiation. Rather than analyzing wavelength-dependent optical properties, the objective was to monitor time-dependent changes in diffuse reflectance resulting from laser-induced alterations in tissue absorption and scattering. These variations reflect changes in tissue hydration, refractive-index matching, and structural integrity associated with thermal damage. Therefore, the proposed DRS framework evaluates the dynamic optical response of burned tissue at a fixed wavelength rather than performing broadband spectroscopic analysis.

The acquired diffuse reflectance signals were stored for subsequent preprocessing and classification. Together with the corresponding thermal images acquired from the same tissue samples under identical experimental conditions, the DRS measurements formed a synchronized dual-modality dataset. This synchronized acquisition enables an objective comparison between thermal imaging and DRS using identical burn conditions and consistent ground-truth labels.

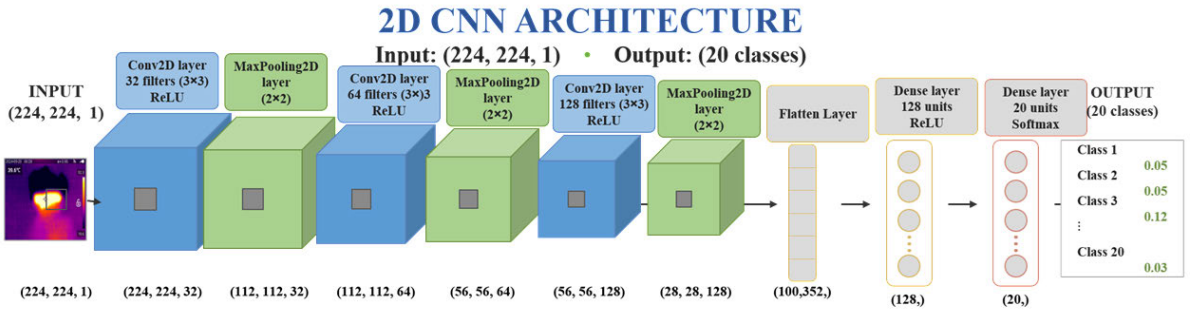
F. DATA COLLECTION

Thermal images and Diffuse Reflectance Spectroscopy measurements were collected from the same ex vivo porcine skin samples immediately after burn induction under identical experimental conditions. For each burn severity level, synchronized measurements were acquired to ensure that both sensing modalities represented the same tissue condition. This synchronized acquisition minimized experimental variability and enabled a fair comparison between thermal imaging and Diffuse Reflectance Spectroscopy.

1) THERMAL IMAGE DATASET

Thermal images were acquired immediately after each burn induction experiment using the infrared camera described in Section III-D. A total of 4,060 thermal images were collected from 20 classes, representing five acquisition conditions for each of the four porcine skin samples, with approximately 200–210 images per class. The images were stored in grayscale format and labeled according to their corresponding burn condition.

Before model training, all images were resized to 224 × 224 pixels and normalized to the [0,1] range. The dataset was partitioned at the specimen level to prevent information leakage and provide an unbiased evaluation of the proposed CNN. Images originating from the same porcine specimen were assigned exclusively to a single data partition; consequently, no specimen contributed images to more than one of the training, validation, or testing datasets. The dataset was then divided into 2,598 training images (64%), 650 validation images (16%), and 812 testing images (20%), while maintaining a balanced class distribution. Data augmentation, including random rotation, horizontal flipping, and translation, was applied only to the training dataset to improve model generalization without introducing augmented samples into the validation or testing datasets. The same data partitioning strategy was maintained throughout model development to ensure a fair and unbiased evaluation of the proposed CNN.



Layer (type)	Filters / Units	Kernel Size	Stride	Padding	Output Shape	Parameters
Input	–	–	–	–	(224, 224, 1)	0
Conv2D	32	(3 × 3)	(1, 1)	same	(224, 224, 32)	320
MaxPooling2D	–	(2 × 2)	(2, 2)	valid	(112, 112, 32)	0
Conv2D	64	(3 × 3)	(1, 1)	same	(112, 112, 64)	18,496
MaxPooling2D	–	(2 × 2)	(1, 2)	valid	(56, 56, 64)	0
Conv2D	128	(3 × 3)	(1, 1)	same	(56, 56, 128)	73,856
MaxPooling2D	–	(2 × 2)	(2, 2)	valid	(28, 28, 128)	0
Flatten	–	–	–	–	(100,352,)	0
Dense	128	–	–	–	(128,)	12,846,144
Dense (Softmax)	20	–	–	–	(20,)	2,580
Total Parameters						12,941,396

ARCHITECTURE SUMMARY

- 3 Convolutional layers (ReLU)
- 3 MaxPooling layers (2×2)
- 1 Flatten layer
- 1 Fully connected layer (128 units, ReLU)
- 1 Output layer (Softmax, 20 classes)

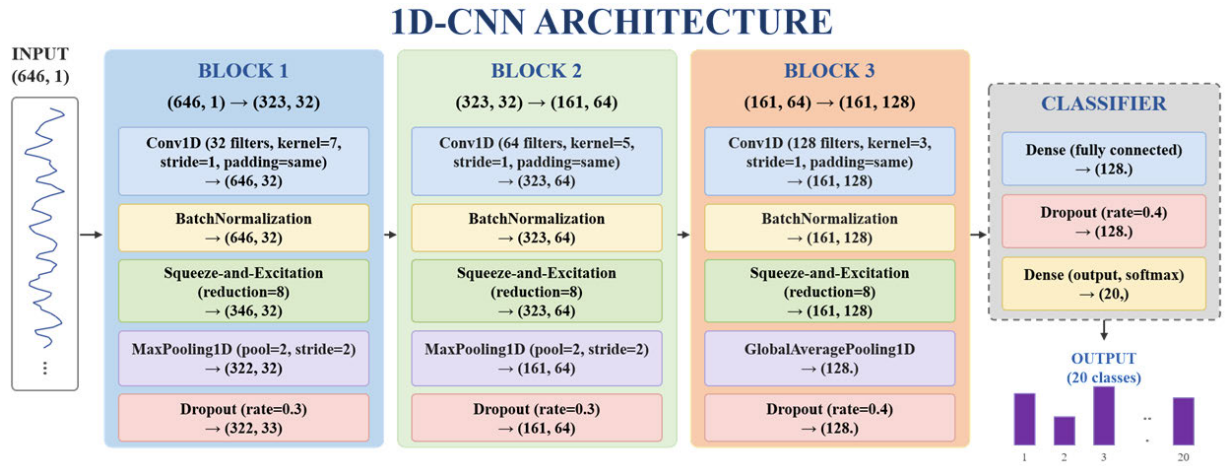
Optimizer : Adam
Loss : Sparse Categorical Crossentropy
Metric : Accuracy

OUTPUT EXAMPLE



All Conv2D layers use padding = 'same' and stride = (1, 1) unless specified.

FIGURE 3. Architecture of the proposed 2D CNN for thermal burn classification, showing the layer configuration, output shapes, and parameter counts.



LEGEND

- Convolution
- Normalization
- Squeeze-and-Excitation
- Pooling
- Dropout
- Fully Connected

OUTPUT SHAPE AFTER EACH STAGE

BLOCK 1	BLOCK 2	BLOCK 3
Input: (646, 1)	Input: (323, 32)	Input: (161, 64)
Conv1D: (646, 32)	Conv1D: (323, 64)	Conv1D: (161, 128)
BN: (646, 32)	BN: (323, 64)	BN: (161, 128)
SE: (646, 32)	SE: (323, 64)	SE: (161, 128)
MaxPool: (323, 32)	MaxPool: (161, 64)	GAP: (128.)
Dropout: (323, 32)	Dropout: (161, 64)	Dropout: (128.)

CLASSIFIER

Dense: (128,)
↓
Dropout: (128,)
↓
Dense (softmax): (20,)

PARAMETERS

Block 1	676
Block 2	11,656
Block 3	29,456
Classifier	19,092
Total	60,880

(60,432 trainable / 448 non-trainable)

All Conv1D layers use bias=True. Padding = same unless specified.

FIGURE 4. Architecture of the proposed 1D-CNN with squeeze-and-excitation blocks for thermal signal classification, showing the layer configuration and output shape after each stage.

2) DIFFUSE REFLECTANCE SPECTROSCOPY DATASET

A total of 1,368 diffuse reflectance measurements were acquired from the same tissue samples under the same experimental conditions and labeled according to the corresponding burn severity level. Each measurement represents the optical response of a single tissue sample at the operating wavelength of 670 nm.

For model development, each diffuse reflectance measurement was represented as a one-dimensional input vector, where each element corresponds to the recorded signal intensity at a specific sampling point. The resulting dataset was organized as a matrix in which each row represents an individual measurement and each column corresponds to a sampled signal value, while pre-

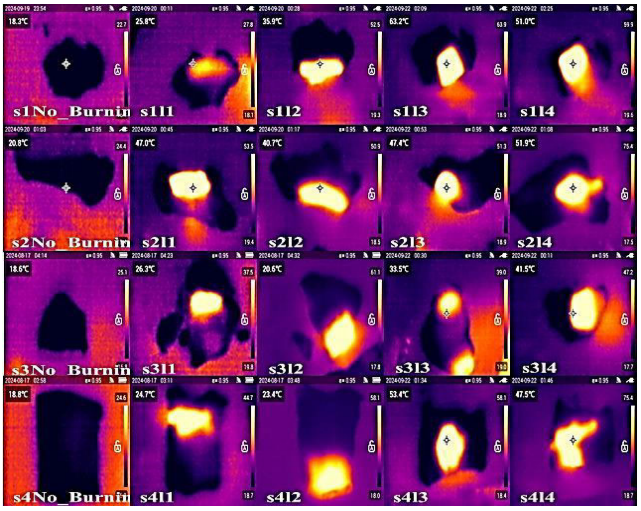


FIGURE 5. Representative thermal images of skin samples before and after burning.

serving the sequential structure required by the proposed 1D-CNN.

To enable an objective comparison between the two sensing modalities, the Diffuse Reflectance Spectroscopy dataset was partitioned into training, validation, and testing subsets using the same evaluation protocol adopted for the thermal image dataset. The partitioning was performed before model training, and the validation and testing datasets were used exclusively for model evaluation. Any preprocessing operations were applied consistently across the dataset, and no augmented or duplicated measurements were included in the testing dataset. This protocol minimizes the risk of information leakage and provides an unbiased assessment of the proposed classification model.

G. BURN SEVERITY CLASSIFICATION USING THERMAL IMAGES

A CNN was developed to classify burn severity from thermal images. CNNs automatically learn hierarchical image features directly from the input data, eliminating the need for handcrafted feature extraction and enabling robust identification of thermal patterns associated with different burn severity levels.

As illustrated in Figure 3, the proposed CNN architecture consists of successive convolutional layers with ReLU activation functions for feature extraction, followed by max-pooling layers that progressively reduce the spatial dimensions while preserving discriminative features. The extracted features are flattened and passed to fully connected layers for burn severity classification. Dropout regularization is incorporated to reduce overfitting, and a SoftMax activation function in the output layer estimates the probability of each burning level.

The model was trained using the Adam optimizer with sparse categorical cross-entropy loss. The implementation details, training configuration, computational environment, and performance evaluation protocol are described in Section III-I.

H. BURN SEVERITY CLASSIFICATION USING DRS

A 1D-CNN was developed to classify burn severity using DRS measurements. Unlike thermal images, which contain two-dimensional spatial information, DRS measurements are represented as one-dimensional sequential signals. A 1D-CNN is therefore well suited for this task because it automatically learns hierarchical spectral features directly from the diffuse reflectance signals without requiring handcrafted feature extraction.

Conventional machine learning methods, such as SVM, Random Forest, XGBoost, and PLS-DA, generally rely on manually engineered features, including principal components, peak ratios, or selected wavelength bands. These features may not fully capture the complex spectral changes associated with progressive thermal damage. In contrast, the convolutional kernels of a 1D-CNN automatically learn local spectral dependencies and feature relationships directly from the input signals, making the model more robust to variations in tissue optical properties. Similar architectures have demonstrated excellent performance in optical spectroscopy and optical fiber sensing applications [63], [64].

As illustrated in Figure 4, the proposed 1D-CNN architecture consists of two Conv1D layers with 64 and 128 filters, respectively. Each convolutional layer is followed by ReLU activation, batch normalization, and max-pooling operations to progressively extract discriminative spectral features. The extracted features are flattened and passed to fully connected layers with 128 and 64 neurons for burn severity classification. Dropout regularization is applied to reduce overfitting, and a SoftMax activation function in the output layer estimates the probability of each burn severity level. The network was trained using the Adam optimizer with categorical cross-entropy loss to optimize the classification performance.

I. MODEL TRAINING AND PERFORMANCE EVALUATION

The proposed CNN and 1D-CNN models were trained and evaluated using the same experimental protocol to ensure a fair comparison between thermal imaging and Diffuse Reflectance Spectroscopy. For each sensing modality, the dataset was divided into training, validation, and testing subsets, as described in Section III-F. The training dataset was used to optimize the model parameters, the validation dataset was used to monitor the learning process and reduce overfitting, and the testing dataset was reserved for the final performance evaluation.

The thermal image classification model was trained using the Adam optimizer with sparse categorical cross-entropy loss for 10 epochs and a batch size of 32. The Diffuse Reflectance Spectroscopy classification model was trained using the Adam optimizer with a cosine-decay-restarts learning-rate schedule (initial learning rate of 1×10^{-3}) and

categorical cross-entropy loss. A batch size of 16 was used, and training was performed for up to 80 epochs with early stopping (patience = 15) to reduce overfitting. The model achieving the best validation performance was selected for the final evaluation.

All experiments were implemented in Python 3.12.13 using TensorFlow 2.20.0 with the Keras 3.13.2 functional API. Data preprocessing and performance evaluation were performed using scikit-learn 1.6.1, while NumPy 2.0.2 and pandas 2.2.2 were used for data processing and analysis. Training and evaluation were conducted on Google Colaboratory running Ubuntu Linux using an NVIDIA Tesla T4 GPU (driver version 580.82.07), ensuring a consistent computational environment for all experiments.

The classification performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, receiver operating characteristic curves, and the corresponding area under the curve. These metrics provide a comprehensive and objective evaluation of both models under identical experimental conditions. Accuracy, precision, recall, and F1-score are defined as

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{1}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$\text{F1} - \text{score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

where TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively.

IV. RESULTS AND DISCUSSION

This section presents and discusses the performance of the proposed deep learning models for burn severity classification using thermal imaging and DRS. The classification results are evaluated using quantitative performance metrics, including accuracy, precision, recall, F1-score, confusion matrices, and ROC analysis. The performance of each sensing modality is first analyzed independently, followed by a comparative discussion to highlight their respective strengths, limitations, and complementary characteristics for burn severity assessment.

A. THERMAL IMAGING RESULTS

1) REPRESENTATIVE THERMAL IMAGES

Figure 5 presents representative thermal images acquired from the four burn severity levels. Progressive changes in the thermal patterns are clearly observed as the burn severity increases. Before burning, the skin surface exhibits a relatively uniform temperature distribution. Following thermal injury, distinct temperature variations develop, with superficial burns showing higher surface temperatures and deeper burns exhibiting progressively lower temperatures due to reduced tissue perfusion and increased structural damage.

Because all thermal images were acquired under identical environmental conditions using the synchronized experimental protocol described in Section III, the observed temperature differences primarily reflect changes in tissue condition rather than variations in the imaging process. These findings demonstrate that thermal imaging provides discriminative spatial information for distinguishing different burn severity

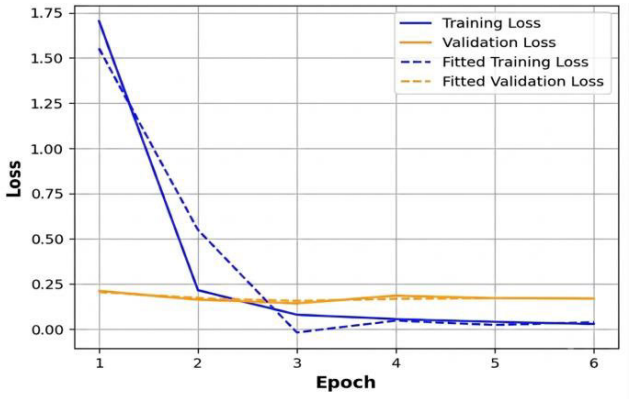


FIGURE 6. Performance plot (training/validation loss).

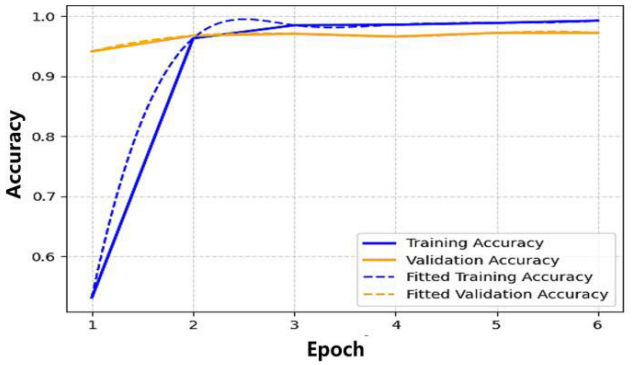


FIGURE 7. Accuracy versus epochs (training/validation accuracy).

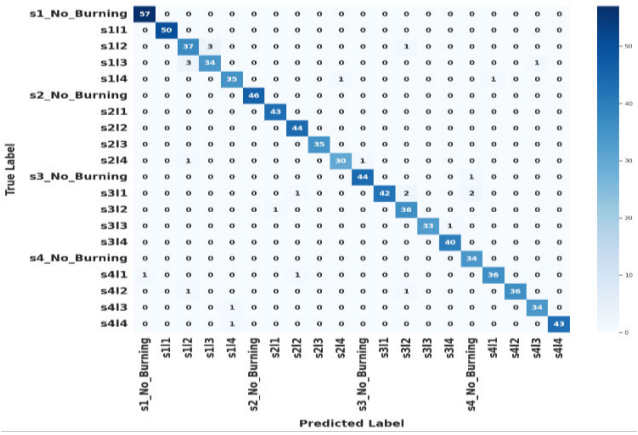


FIGURE 8. The confusion matrix for thermal imaging dataset.

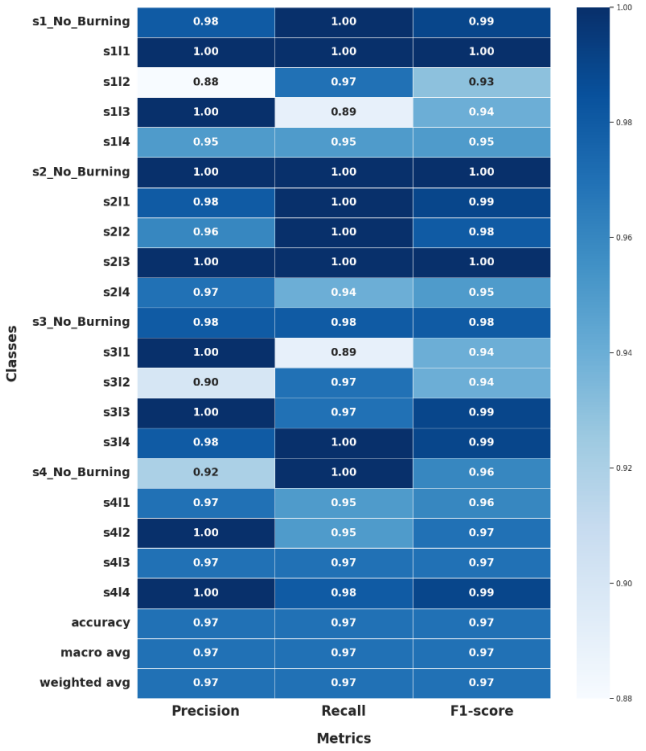


FIGURE 9. Classification report heatmap.

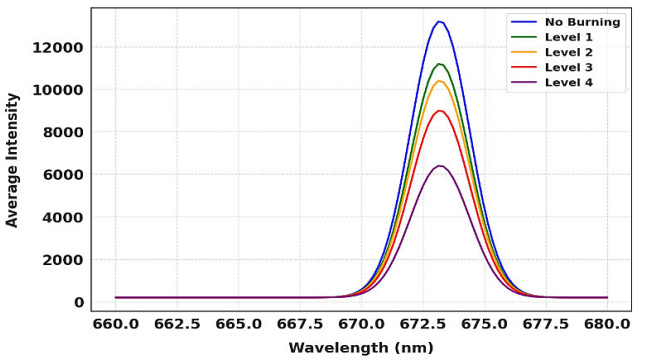


FIGURE 10. Spectral intensity variations across different burn severity levels for the same sample.

levels and supports its use as an effective input modality for automated burn severity classification.

2) CNN TRAINING AND VALIDATION PERFORMANCE

The learning behavior of the proposed CNN model is presented in Figures 6 and 7. Figure 6 illustrates the training and validation loss during model training, whereas Figure 7 presents the corresponding training and validation accuracy.

As shown in Figure 6, both the training and validation loss decrease rapidly during the initial training epochs before gradually converging to stable values. The final training and validation loss reached approximately 0.04 and 0.10, respectively, indicating effective optimization and stable convergence throughout the training process. The close

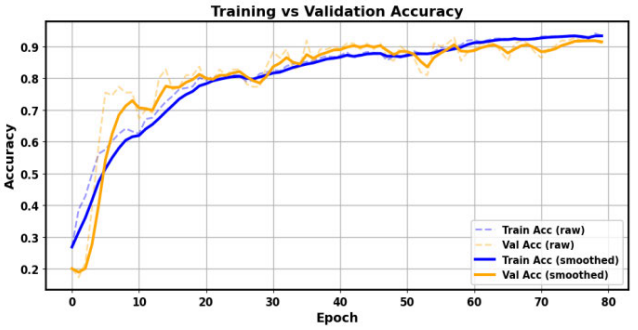


FIGURE 11. Training and validation accuracy over epoches.

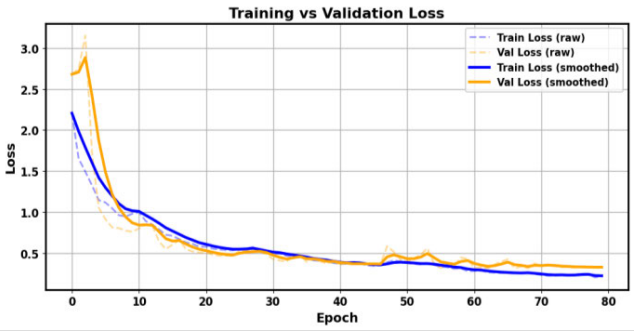


FIGURE 12. Training and validation losses over epoches.

agreement between the two curves suggests good generalization with no evidence of significant overfitting.

The corresponding accuracy curves are presented in Figure 7. The training accuracy increased steadily and reached approximately 99%, while the validation accuracy converged to approximately 97%. The small difference between the two curves demonstrates that the proposed CNN generalized well to previously unseen data while maintaining stable learning throughout the training process.

Overall, the training and validation results demonstrate that the proposed CNN effectively learned discriminative thermal features associated with different burn severity levels. The combination of low loss values and high classification accuracy confirms that the selected network architecture and training strategy are well suited for thermal image-based burn severity classification.

3) CONFUSION MATRIX ANALYSIS

The confusion matrix shown in Figure 8 evaluates the classification performance of the proposed CNN across the 20 sample–burn-level classes. The matrix exhibits strong diagonal dominance, indicating that the vast majority of thermal images were correctly classified. Most classification errors were confined to adjacent burn severity levels within the same skin sample, while no confusion was observed between non-adjacent burn levels (e.g., Level 1 and Level 4), demonstrating that the model effectively preserved the progression of burn severity.

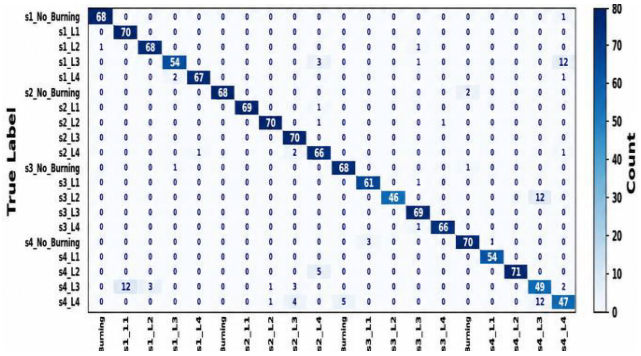


FIGURE 13. Confusion matrix for spectral dataset.

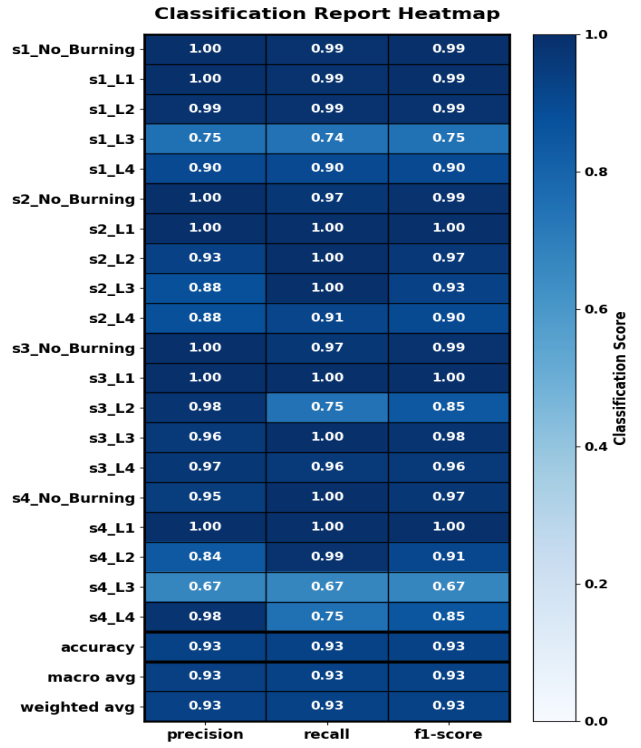


FIGURE 14. Classification report heatmap showing precision, recall, and F1-score for each class.

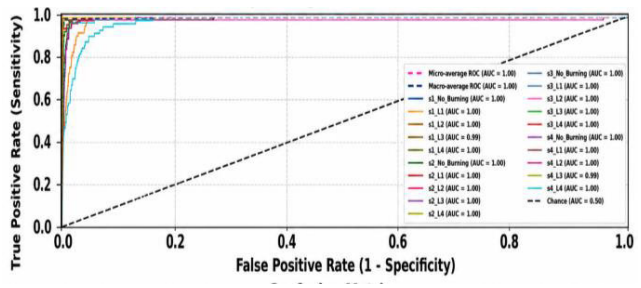


FIGURE 15. Multi-Class ROC curves illustrating the classification performance across different classes, with AUC values representing the model's ability to separate each class effectively.

Several classes were classified without error. For example, all 57 images of s1_No_Burning and all 50 images of s1_L1 were correctly identified. Similarly, the classes s2_No_Burning, s3_L3, and s4_L4 achieved perfect classification. The remaining errors primarily occurred between neighboring burn levels, such as three s1_L3 samples being classified as s1_L2 and one s4_L2 sample being classified as s4_L3. These results indicate that the remaining classification errors arise from the gradual transition between adjacent burn severities rather than from a failure of the proposed CNN to distinguish different burn categories.

4) CLASSIFICATION PERFORMANCE

The quantitative classification results are summarized in Figure 9. The proposed CNN achieved an overall accuracy of 97.17%, with a weighted precision of 97.30%, weighted recall of 97.17%, and weighted F1-score of 97.17%. In addition, the macro-average F1-score reached 97.03%, demonstrating consistent classification performance across all 20 burn classes rather than being dominated by a small number of categories.

Most burn classes achieved precision, recall, and F1-score values close to 1.00, indicating excellent discrimination between burn severity levels. The highest classification performance was observed for s1_No_Burning, which achieved a precision of 0.98, recall of 1.00, and F1-score of 0.99. Slightly lower performance was observed for s1_L2, where the precision decreased to 0.88, and for s3_L1, where the recall was 0.89, reflecting a small number of misclassifications between adjacent burn levels. Despite these minor errors, the consistently high evaluation metrics confirm that the proposed CNN accurately identifies burn severity from thermal images and provides a reliable framework for automated burn assessment.

B. SPECTROSCOPIC RESULTS

1) SPECTRAL CHARACTERISTICS

Figure 10 presents representative DRS signals acquired from the different burn severity levels. Progressive changes in the diffuse reflectance signals are observed as burn severity increases, indicating that thermal injury alters the optical properties of the tissue. These variations arise from changes in tissue absorption and scattering associated with structural damage, dehydration, and modifications in tissue composition.

The representative signals shown in Figure 10 are consistent with the overall trends observed across all four porcine skin specimens. Across all specimens, thermal injury produced progressive changes in the diffuse reflectance signals with increasing burn severity. Although minor differences in absolute signal intensity were observed because of natural biological variability, including differences in tissue thickness, hydration, and optical properties, the overall response pattern remained consistent. These findings indicate that the measured optical responses are primarily associated with burn-induced tissue changes rather than specimen-specific

TABLE 2. Comparative analysis of CNN models for burn severity classification: Thermal imaging vs. Spectroscopic data.

Aspect	Thermal Imaging Modality	Spectroscopic Modality
Data Type & Source	2D spatial thermal images (infrared) captured with a UTi120TN thermal camera.	1D diffuse reflection spectral vectors captured with an Aurora 4000 fiber-optic spectrometer.
Experimental Stimulus	Controlled exposure to a 670 nm laser; burn severity defined by exposure duration (L1–L4).	Same controlled burn induction protocol with spectra captured at each stage.
Dataset Size & Structure	4060 grayscale images (320×240 pixels), 20 classes.	20 CSV files (4 skin samples × 5 burn levels); 1,368 total readings after NaN/outlier cleaning; raw spectral length 3,230 points, downsampled to 646
Data Preprocessing	Resizing to 224×224, normalization, 64/16/20 split.	Per-file NaN/outlier cleaning, 5-point mean-pooling downsampling, 1st/99th-percentile clipping, RobustScaler normalization, SMOTE-style same-class oversampling (training folds only), stratified 5-fold cross-validation.
CNN Architecture	2D CNN with convolutional layers (32, 64, 128 filters) and softmax output.	1D CNN: three Conv1D blocks (32→64→128 filters, kernels 7/5/3) with batch normalization, squeeze-and-excitation attention, max-pooling and dropout (0.3–0.4); L2-regularized dense layer (128 units) before softmax output.
Training Configuration	Adam optimizer, sparse categorical cross-entropy, 10 epochs, batch size 32.	Adam optimizer with cosine-decay-restarts schedule (initial LR 1e-3) and gradient clipping; categorical cross-entropy; batch size 16; up to 80 epochs with early stopping (patience 15); fresh model per CV fold.
Key Performance Metrics	Accuracy 97.17%, Precision 97.30%, Recall 97.17%, F1-score 97.17%.	Out-of-fold accuracy 93.0% (n=1,368); macro/weighted avg precision, recall, F1 all 0.93; AUC = 1.00 for 17/20 classes, 0.98–0.99 for the remaining three (s1_L3, s3_L2, s4_L3).
Conclusion & Synergy Potential	Excellent spatial assessment with minor adjacent-level misclassification.	Strong biochemical differentiation; complementary to thermal imaging.

characteristics, demonstrating the reproducibility of the proposed controlled experimental framework.

Although the measurements were acquired at a fixed wavelength of 670 nm, the temporal diffuse reflectance signals exhibited distinct response patterns for different burn severity levels. The synchronized acquisition protocol described in Section III ensured that all measurements were obtained under identical experimental conditions, allowing the observed differences to be attributed primarily to burn-induced tissue changes rather than measurement variability. These results demonstrate that DRS provides complementary physiological information for burn severity classification.

2) 1D-CNN TRAINING AND VALIDATION PERFORMANCE

The learning behavior of the proposed 1D-CNN model is illustrated in Figures 11 and 12. Figure 11 presents the training and validation loss during model training, whereas Figure 12 shows the corresponding training and validation accuracy.

As shown in Figure 11, the training accuracy increased steadily throughout the learning process and converged to approximately 95%, whereas the validation accuracy stabilized at approximately 93%. The close agreement between the two curves indicates that the model generalized well to unseen spectral measurements and achieved stable learning without significant overfitting.

The corresponding loss curves in Figure 12 decreased consistently during training before converging to stable values. Both the training and validation loss followed similar trends with only a small separation between the curves, demonstrating effective optimization and stable convergence throughout the training process. These learning curves confirm that the proposed 1D-CNN successfully learned discriminative spectral features suitable for burn severity classification.

3) CONFUSION MATRIX ANALYSIS

The confusion matrix presented in Figure 13 provides a detailed evaluation of the proposed 1D-CNN across the 20 sample–burn-level classes. The matrix exhibits strong diagonal dominance, indicating that the majority of spectral measurements were correctly classified. The most notable classification errors occurred among the deeper burn severity levels, particularly where s4_L3 was misclassified as s1_L3 in 12 instances. Additional confusion was observed between s3_L2 and s4_L2, and between s4_L4 and s4_L2, reflecting the increased similarity of spectral signatures in advanced burn stages.

Despite these challenging cases, the overall confusion matrix demonstrates strong classification performance. The remaining misclassifications were largely confined to neighboring burn severity levels, indicating that the proposed 1D-CNN effectively captured the progressive optical changes associated with increasing tissue damage while maintaining high discrimination across most classes.

4) CLASSIFICATION PERFORMANCE

The quantitative classification results are summarized in Figure 14. The proposed 1D-CNN achieved an overall classification accuracy of 93.0%, while the macro-average and weighted-average precision, recall, and F1-score all reached 0.93, demonstrating balanced performance across all 20 burn classes.

Most burn classes achieved precision, recall, and F1-score values close to 1.00, indicating excellent separability between burn severity levels. For example, s1_No_Burning, s1_L1, s1_L2, s2_L1, s3_L1, and s4_L1 all achieved F1-scores

at or near 1.00. The lowest classification performance was observed for s4_L3, with a precision, recall, and F1-score of 0.67, followed by s1_L3, which achieved an F1-score of 0.75. In addition, s3_L2 achieved a precision of 0.98 but a recall of 0.75, indicating that a small number of samples were classified as neighboring burn severity levels.

5) ROC ANALYSIS

The ROC curves shown in Figure 15 further evaluate the discrimination capability of the proposed 1D-CNN. The majority of the 20 burn classes achieved an AUC of 1.00, demonstrating excellent class separability and high classification confidence. Furthermore, both the micro-average and macro-average ROC curves also achieved an AUC of 1.00, indicating consistently strong performance across the complete dataset.

Only three classes exhibited slightly lower AUC values, with s1_L3 achieving an AUC of 0.99, and both s3_L2 and s4_L3 achieving AUC values of 0.98. These slight reductions are consistent with the confusion matrix analysis and reflect the greater similarity between adjacent high-severity burn levels. Nevertheless, the ROC analysis confirms the robustness and reliability of the proposed 1D-CNN for DRS-based burn severity classification.

C. COMPARATIVE ANALYSIS

The quantitative results demonstrate that both thermal imaging and Diffuse Reflectance Spectroscopy (DRS) are effective for automated burn severity classification under the same controlled experimental framework. The synchronized acquisition of thermal images and DRS measurements from the same tissue samples under identical burn conditions enables an objective comparison between the two sensing modalities.

The CNN developed for thermal imaging achieved an overall classification accuracy of 97.17%, with a weighted precision of 97.30%, weighted recall of 97.17%, and weighted F1-score of 97.17%. The 1D-CNN developed for DRS achieved an overall classification accuracy of 93.00%, with macro-average and weighted-average precision, recall, and F1-score values of 0.93. Under the adopted evaluation protocols, the thermal imaging model achieved higher classification performance than the DRS model. However, because the two models were evaluated using different validation strategies, no formal statistical significance test was performed. Therefore, the observed performance difference should be interpreted as a quantitative comparison rather than evidence of statistical superiority.

The confusion matrix analysis further supports these findings. The thermal imaging model produced fewer misclassifications, with most errors occurring only between adjacent burn severity levels. In contrast, the DRS model exhibited greater confusion among neighboring burn classes because progressive thermal damage produces similar tissue absorption and scattering characteristics, making adjacent severity levels more difficult to distinguish. Nevertheless, both models showed minimal confusion between non-adjacent burn severity levels, indicating that they effectively preserved the overall burn severity progression.

Although thermal imaging achieved higher standalone classification performance, the two sensing modalities provide complementary diagnostic information. Thermal imaging characterizes surface temperature distributions associated with tissue perfusion and thermal damage, whereas DRS measures changes in tissue absorption and scattering that reflect physiological and structural alterations. Consequently, the two techniques should be regarded as complementary rather than competing approaches for burn severity assessment.

Table 2 summarizes the quantitative comparison between thermal imaging and DRS using the proposed synchronized dual-modality dataset and controlled experimental framework. Thermal imaging achieved higher overall performance across the evaluated metrics, whereas DRS provided complementary physiological information that cannot be obtained from thermal images alone. These findings establish a reliable benchmark for future multimodal deep learning methods that integrate both sensing modalities for automated burn severity classification.

V. CONCLUSION

This study presented a controlled experimental framework for synchronized burn severity classification using thermal imaging and Diffuse Reflectance Spectroscopy acquired from the same ex vivo porcine skin samples under identical experimental conditions. The framework enabled the development and objective evaluation of separate deep learning models for the two sensing modalities.

The experimental results demonstrated that both sensing modalities achieved high classification performance. The thermal imaging model achieved an overall classification accuracy of 97.17%, with a weighted precision of 97.30%, weighted recall of 97.17%, and weighted F1-score of 97.17%. The DRS model achieved an overall classification accuracy of 93.00%, with macro-average and weighted-average precision, recall, and F1-score values of 0.93. Overall, thermal imaging achieved higher standalone classification performance than DRS under the same experimental conditions.

The main contribution of this work is the development of a synchronized dual-modality dataset and experimental framework that enables a fair and objective comparison between thermal imaging and Diffuse Reflectance Spectroscopy using the same tissue samples, burn conditions, and ground-truth labels. This framework provides a reliable benchmark for future studies on automated burn severity classification and multimodal burn assessment.

Future work will focus on multimodal deep learning models that integrate thermal imaging and Diffuse Reflectance Spectroscopy, together with larger datasets, histopathological validation, and clinical studies to further improve classification performance and support clinical translation.

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